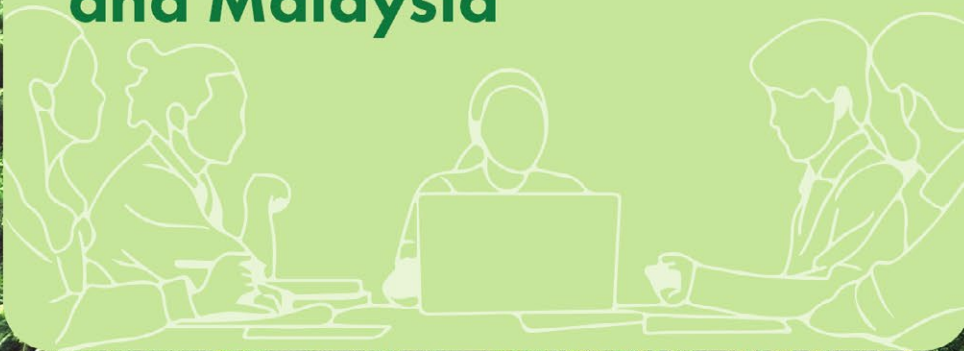


Inclusive Traceability Systems for EUDR Readiness for Independent Smallholders in Indonesia and Malaysia



Implementation Partners:



Solidaridad

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Glossary of Terms

APIs	: Application Programming Interfaces
APL	: Area Penggunaan Lain
CASCADE Project	: an EU-funded project in Indonesia: Michelin and Porsche AG support farmers in sustainable natural rubber farming, protecting biodiversity and livelihoods.
CoC	: Chain of Custody
DDS	: Due Diligence Statement (required under EUDR)
DDS Readiness	: Preparedness of operators/suppliers to submit a compliant Due Diligence Statement under EUDR
DPGs	: Digital Public Goods. Open-source digital solutions that adhere to privacy and applicable standards and contribute to the Sustainable Development Goals (SDGs)
e-MSPO	: Electronic Malaysian Sustainable Palm Oil - A digital system supporting certification and traceability under the MSPO scheme
e-STDB	: Electronic Surat Tanda Daftar Budidaya - Electronic Registration Certificate for Plantation Cultivation (Indonesia)
EUDR	: European Union Deforestation Regulation (Regulation (EU) 2023/1115 on deforestation-free products)
FAO's Whisp	: What is in that Plot? - A geospatial assessment tool developed by FAO to assess deforestation risk at plot level
Farmer IDs	: Unique Farmer Identification Numbers used for traceability and compliance
FELDA	: Federal Land Development Authority (Malaysia)
FFB	: Fresh Fruit Bunches (oil palm)
GAP	: Good Agricultural Practices
GeoSawit Platform	: Malaysia geospatial platform supporting palm oil traceability and smallholder mapping
ICS	: Internal Control System (used in group certification schemes such as RSPO or MSPO)
HGB	: Hak Guna Bangunan
HGU	: Hak Guna Usaha
IDH	: Inisiatif Dagang Hijau
JRC	: Joint Research Centre (European Commission)
JSON-based APIs	: APIs that use JavaScript Object Notation (JSON) as the data exchange format
KKSS Cooperative	: Koperasi Kakao Kerta Semaya Samaniya, based in Jembrana District, Bali
KPIs	: Key Performance Indicators (e.g., mapped plots, verified suppliers, traceable volumes)
LGM	: Lembaga Getah Malaysia - Malaysian Rubber Board (MRB)
MPOB	: Malaysian Palm Oil Board
MPOCC	: Malaysian Palm Oil Certification Council
MSNR	: Malaysian Sustainable Natural Rubber (certification scheme and regulatory framework)
MSNR Trace	: Traceability system under the MSNR certification framework
MSPO 2.0	: Malaysian Sustainable Palm Oil standard updated version
MSPO Trace	: The digital traceability and transaction platform supporting MSPO-certified supply chains, known as e-MSPO
MYSA	: Malaysian Space Agency - in Malaysia, is used to cross reference high-resolution imagery to verify compliance with December 2020 cut-off date
NFI	: National Forest Inventory
NIK	: Nomor Induk Kependudukan - Indonesian National Identity Number
OFI	: Olam Food Ingredients
One Map Policy	: Indonesian government initiative (Kebijakan Satu Peta) to harmonize geospatial data into a single reference map
Open Foris Ground	: An open-source, map-first data collection platform co-developed by FAO and Google
PAT-G	: Pas Kebenaran Getah - Rubber Transaction Permit (Malaysia)
PAT-G Card	: Rubber Smallholder Permit
PIPOC 2025	: MPOB International Palm Oil Congress and Exhibition
PPI Compact	: Public-Private Investment Compact
RRIM GeoRubber	: Geospatial traceability and mapping platform developed by Malaysian Rubber Board
RRIMniaga	: Digital trading platform for Malaysian rubber transactions under MRB/LGM
RSPO	: Roundtable on Sustainable Palm Oil
RTD	: Regional Technical Dialogue
SDOs	: Sourcing Development Officers
SHM	: Sertifikat Hak Milik - Freehold Land Ownership Certificate (Indonesia)
SIPERIBUN	: Sistem Perizinan Berusaha Perkebunan - Indonesia's Online Plantation Licensing System
SI-ISPO	: Sistem Informasi ISPO - Information system supporting Indonesian Sustainable Palm Oil certification
SIMS	: Sawit Intelligent Management System
SIMONTANA	: Sistem Monitoring Hutan Nasional - Indonesia's National Forest Monitoring System
SKN	: Sistem Kebolehejekan Nasional - National Traceability System (Malaysia)
SKT	: Surat Keterangan Tanah - Land ownership/land statement letter issued by the village head (Indonesia)
SMEs	: Small and Medium-sized Enterprises
SPOC Model	: Sustainable Palm Oil Cluster Model
STDB	: Surat Tanda Daftar Budidaya - Registration Certificate for Plantation Cultivation (Indonesia)
TUNAS Officer	: Extension officer under MPOB's TUNAS (Tunjuk Ajar dan Nasihat Sawit) program providing advisory services to oil palm smallholders
VSA	: Verified Sourcing Area

Executive Summary

Convened by a consortium consisting of the Cocoa Sustainability Partnership (CSP), Partnership for Sustainable Agriculture (PISAgro), Solidaridad Indonesia and Malaysia, and the Tropical Forest Alliance (TFA), this publication presents the outcomes of Regional Technical Dialogue #4 on strengthening practical traceability solutions for compliance and market competitiveness.

The report was specifically designed to address traceability gaps, explore the role of Digital Public Goods, and outline the recommendation for breakthroughs required to maintain smallholder inclusion. Ultimately, this publication serves as a practical blueprint for governments and the private sector to bridge the divide between farm-gate realities with smallholders, and EU market requirements.

This analysis reveals a sharp divide in regional readiness. Malaysia possesses a robust institutional structure, though challenges remain in fully operationalizing the National Traceability System (SKN) for independent smallholders. In contrast, Indonesia's decentralized landscape makes closing the middle gap more difficult. Success there depends on innovations like digital commodity identities (palm, rubber, and cacao) to ensure traceability is not lost as products move from farmers to intermediaries.

The findings presented here that technology alone provides lack of sufficient solution. It is argued that the industry must pivot from isolated plot-mapping toward a model of Verified Aggregation. By focusing on three critical breakthroughs, the current implementation challenges can be resolved: (1) the utilization of Digital Public Goods (DPGs) such as Whisp and Open Foris Ground to ensure cross-border system interoperability; (2) specifically for the independent smallholders and the need to provide individual-farm level data through the Verified Sourcing Areas (VSA) or Jurisdictional Umbrellas; and (3) the formalization of informal collectors into Verified Aggregators to ensure the preservation of farmer IDs (e-STDB or PAT-G (Rubber Transaction Permit) enables or supports traceability at the point of transaction.

The report concludes with a recommendation designed to facilitate this transition. Recognizing that national dashboards in Indonesia is currently in a state of development, it is recommended that this platforms evolve from static data repositories into dynamic validation engines capable of supporting real-time updates. For the private sector, the necessity of moving beyond proprietary, siloed applications is emphasized. Instead, investment is required in interoperable systems that allow smallholders to be integrated into the global market as investable clusters rather than high-risk outliers.

Source: iStock

Introduction:

the EUDR, National Traceability Systems and Smallholders Inclusion

The EU Market Compliance, National Traceability Architectures & Value Chain actors and the Smallholder Context

The EU Deforestation Regulation (EUDR), adopted in 2023 with application beginning in December 2026 for large operators and June 2027 for mid-scale companies, establishes a new global benchmark for commodity trading. It marks a paradigm shift from voluntary standards to mandatory company due diligence beyond certification. Exported goods can no longer simply be claimed as sustainable; instead, traders and operators placing commodities on the EU market are now required to ensure their goods are supported by verifiable geospatial evidence and proof that their entire supply chain is deforestation-free and legally produced.

However, for operators to fulfil this mandate, they must navigate a production landscape dominated by independent smallholders rather than large estates. The global supply of forest-risk commodities is highly fragmented, making it difficult to isolate specific smallholders without incurring significant volume losses or excluding producers whose geolocations are tied to first-hand commodity supplies. In the Palm oil sector smallholders manage approximately 40%¹ of the total planted area globally. In Malaysia, while organized schemes like FELDA are prominent, independent smallholders contribute a significant portion (approximately 16%) of national palm oil production. Due to high operating costs and limited logistical access, these farmers typically rely on intermediaries who collect the fruits, rather than selling directly to mills. In contrast, smallholders are the primary drivers of palm oil: they account for 85-90%² of production for both Indonesia and Malaysia, cacao smallholders account for nearly 90-95%³ of the global supply.

¹ International Rubber Study Group (IRSG), Statistical Summary (2024); Indonesian Rubber Council (Gapkindo); Malaysian Rubber Board (LGM); and International Cocoa Organization (ICCO), Quarterly Bulletin of Cocoa Statistics, 2024.

² Lestari, D., & Oktavilia, S. (2020). Analysis of Palm oil price in Southeast Asia. AFEBI Economic and Finance Review, 5(02), 63. <https://doi.org/10.47312/aefer.v5i02.494>

³ Fatkurrohim, Purwono, & Kovshov. (2025). The impact of Indonesia's cocoa downstream policy on the derivative product competitiveness, export specialization, and farmers benefit. HABITAT, 36(2), 154-165.

While regulations hold operators legally liable, the informal and fragmented nature of raw material sourcing creates a significant compliance challenge. Unlike digitized corporate plantations, independent smallholders operate on small, dispersed plots, reaching the market through a multi-tiered network of middlemen. This complex aggregation often obscures origin data before the product ever reaches the processing facility. To bridge this gap and meet EUDR requirements, operators must implement a rigorous three-phase compliance process across regulated commodities. This publication focuses specifically on the three primary exports from Indonesia and Malaysia: palm oil, rubber, and cocoa.

Given the gap between regulatory theory and the reality of current sourcing—which relies on informal and oral agreements and cash transactions—the transition to

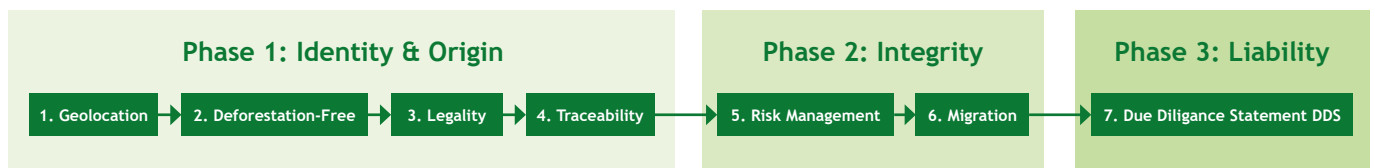
Phase 1 creates a data blind spot. Without a digital record at the point of collection, liability remains with the Operator, yet the evidence stays trapped in the informal tier of the supply chain.

Phase 1: Identity & Origin (Data Gathering): Geolocation (polygons for plots >4ha), deforestation-free verification (post-Dec 2020), legality (national law adherence), and chain-of-custody maintenance.

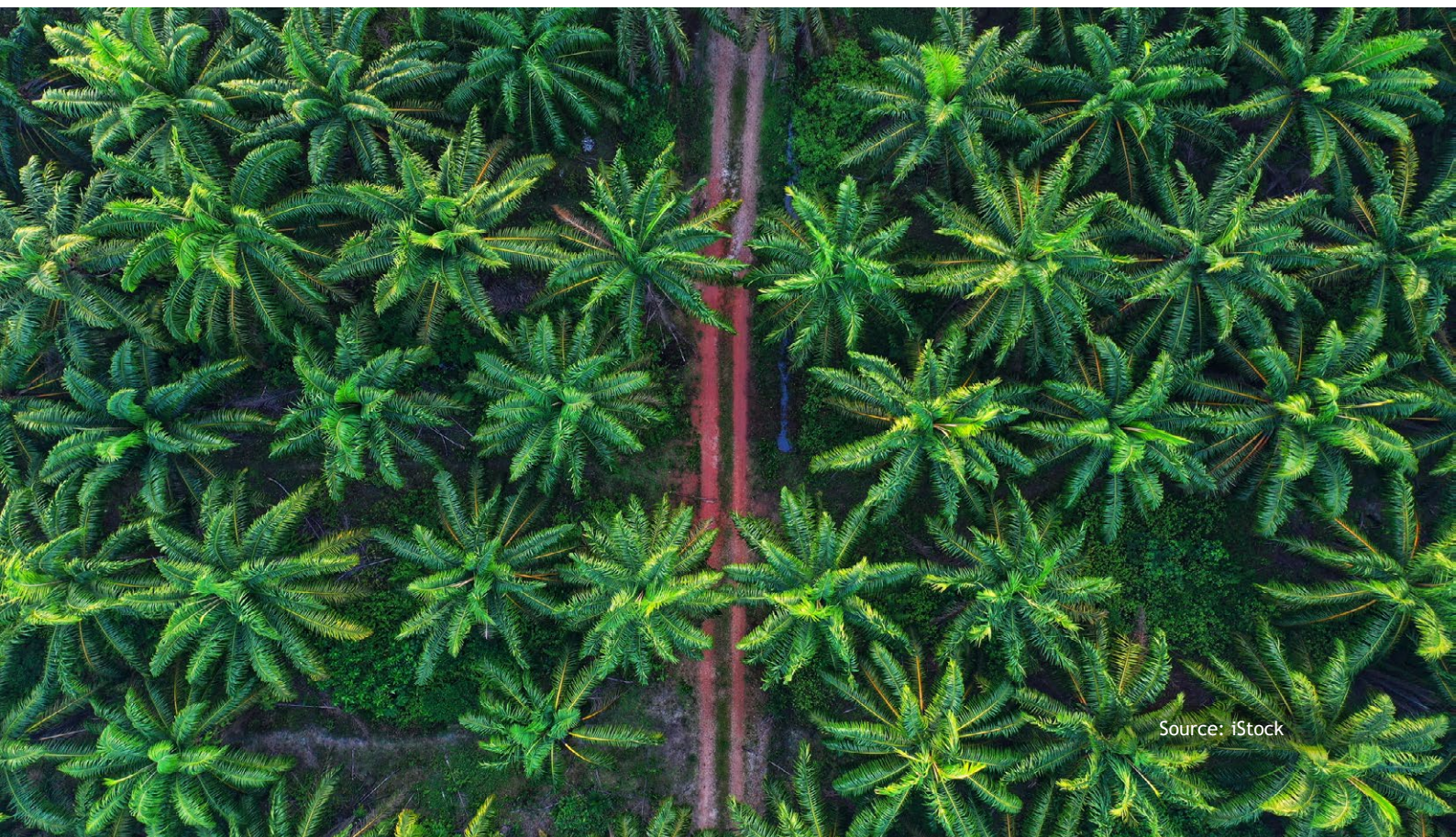
Phase 2: Integrity & Management (Risk Validation): Identifying supply chain gaps and performing on-the-ground mitigation for high-risk areas.

Phase 3: Liability (Compliance Declaration): Submitting the Due Diligence Statement (DDS) to declare accuracy and assume legal liability.

Figure 1 - Distinct Phases of Commodity Digital Products



Source: Synthesized by the author (2026), based on the requirements of Article 9 (Information), Article 10 (Risk Assessment), and Article 12 (Due Diligence Statement) of Regulation (EU) 2023/1115.



The success of compliance to the EUDR relies on transfer of data from farm to trader and operator, yet this flow is currently obstructed by a significant readiness gap among upstream actors. Ideally, smallholders equipped with verifiable geolocation and legal registration—such as the STDB (smallholder cultivation registration) in Indonesia or the MPOB (Malaysian Palm Oil Board) License (for palm) and PAT-G Card (Rubber Transaction Authority Permit) in Malaysia—provide this digital identity during the transaction. In Indonesia, accelerating e-STDB enrolment and geolocation mapping for independent growers is essential to enable farmers to maintain market access for palm oil, cacao and rubber. In Malaysia, while MPOB licensing is robust, gaps persist in the rubber sector and in the independent palm oil sector, where critical traceability data often stalls at the dealer level due to a lack of incentives for intermediaries to participate.

Both nations have launched initiatives to bridge these gaps, evolving from national sustainability certifications toward centralized data architectures. In Malaysia, this is moving toward a unified digital ecosystem aimed at vertical integration, though the full operationalization of systems like the SKN and smallholder inclusion remain ongoing. Similarly, Indonesia is navigating its digital transition through the SIPERIBUN dashboard within the framework of Plantation Law 39/2014, which provides guidelines on data confidentiality for plantation actors (*Pelaku Usaha Perkebunan*)

While private sector traceability systems have reached high levels of technological sophistication, they have also highlighted a vital need for structural interoperability. Before state-level systems were fully realized, the deployment of proprietary platforms by multinational companies provided early momentum for transparency. Moving forward, the priority is to bridge these disconnected ‘digital silos’ to ensure that suppliers—particularly dealers and mills—can operate within a more streamlined and unified reporting environment.

Establishing a systemic foundation of interoperable tools and unified identifiers transforms data collection from a repetitive requirement into a strategic

asset. This alignment not only optimizes operational efficiency but also safeguards the integrity of the Due Diligence Statement (DDS). By harmonizing disparate data formats, operators can more confidently provide the high-fidelity evidence required for compliance, effectively mitigating legal risks such as data mismatch or volume double-counting while strengthening the overall resilience of the value chain.⁵

Suppliers, particularly dealers and mills, are currently operated in disconnected silos required to report across multiple, conflicting companies with their own system and technology - that later feed to the traders’ platforms. This lack of data alignment impacts operational efficiency and administrative streamlining, demonstrating that without a systemic foundation of interoperable tools and unified identifiers, traceability remains a resource-intensive process. Establishing this technical coherence is essential to transform data collection into a value-adding mechanism that enhances the resilience and transparency of the regional value chain. For an operator to submit a DDS, they must currently reconcile disparate data formats from multiple private dashboards, increasing the risk of legal liability due to data mismatch or double-counting of volumes.

The Pathway of Decentralized Aggregation in Indonesia

To navigate its vast archipelagic geography, Indonesia has moved away from a monolithic database toward a decentralized model of interoperability anchored by the proposed National Dashboard. This system is currently being developed is designed to aggregate and reconcile data from existing legal entities rather than replace them.

For the palm oil sector, the dashboard triangulates data from established government platforms—SIPERIBUN (corporate licensing), e-STDB (smallholder cultivation licensing), and SI-ISPO (sustainability certification)—alongside private sector inputs⁶. However, the rubber

⁴ Republic of Indonesia. (2014). *Undang-Undang Republik Indonesia Nomor 39 Tahun 2014 tentang Perkebunan* [Law of the Republic of Indonesia Number 39 of 2014 concerning Plantations]. State Secretariat.

⁵ FGD PS

and cocoa sectors face more precarious challenges. Traders in these commodities can collect geolocation data yet they still face the lack of land proof of legality. Consequently, companies are unable to generate legality data internally and rely almost on the government-issued e-STDB as their primary compliance anchor. Unlike the palm oil sector, which is bolstered by large corporate estates and vertical integration, the rubber and cacao sectors are overwhelmingly dominated by independent smallholders, shifting the legality verification directly onto state systems.

This reliance exposes a severe structural weakness. First, field implementation is critically low—estimated at just 5%—leaving the vast majority of the supply base invisible to formal systems. Second, and more damaging to long-term traceability, is the lack of dynamic updates. Currently, the e-STDB functions as a static snapshot rather than a living registry, with no active mechanism to record changes in land tenure, crop profiles, or deforestation events after the initial certificate is issued.

This data stagnation creates a legality blind spot, making it operationally challenging for traders to verify goods against a database that may be effectively obsolete. Without a dynamic registry, the link between the physical commodity and its legal origin is broken. To bridge this gap, the National Dashboard must evolve from a repository into a dynamic validation engine. By integrating real-time satellite monitoring and periodic digital check-ins, the system can refresh the legality status of a plot without requiring a full e-STDB re-registration.

The proposed National Dashboard attempts to act as the validation gate⁶ for this fragmented governance, where jurisdiction over plantation legality, forest cover, and spatial planning is split across the Ministry of Agriculture (SIPERIBUN), the Ministry of Environment and Forestry (forest layers), ATR/BPN (land titles), and local district licensing bodies. By consolidating these disparate datasets into a single export-facing interface, the dashboard aims to eliminate duplication and provide the consistent outputs operators need to prepare EUDR Due Diligence Statements.

To ensure data integrity as goods move from farmers to collectors, the framework establishes a gatekeeper validation protocol. Departing from Malaysia's model of direct digital recording, Indonesia strategy is to require processing mills – which utilize a variety of proprietary Enterprise Resource Planning (ERP) systems and third-party traceability platforms (such as Koltiva or so forth) – as the primary verification point, whereas factories act as mandatory checkpoints, required to report the e-STDB IDs of incoming supplies. This mechanism forces the network of informal collectors to preserve the original farmer's identity. By reconciling incoming volumes against the registered e-STDB quotas at the factory gate, the system aims to filter unverified goods out of the export stream.

Geospatially, One Map Policy as mandated by Presidential Regulation No 23 of 2021, to resolve the fractured supply chain. While the technical integration of thematic maps has faced delays, it remains the state-level tool for reconciling overlapping land-tenure claims—most notably the ongoing disparities between Ministry of Agriculture (MoA) plantation permits and Ministry of Environment and Forestry (MoEF) forest zone designations. The objective of this map is to perform as a validator where data—whether from a palm estate or a cocoa garden—is validated against the One Map corridor and presented as a unified compliance package.

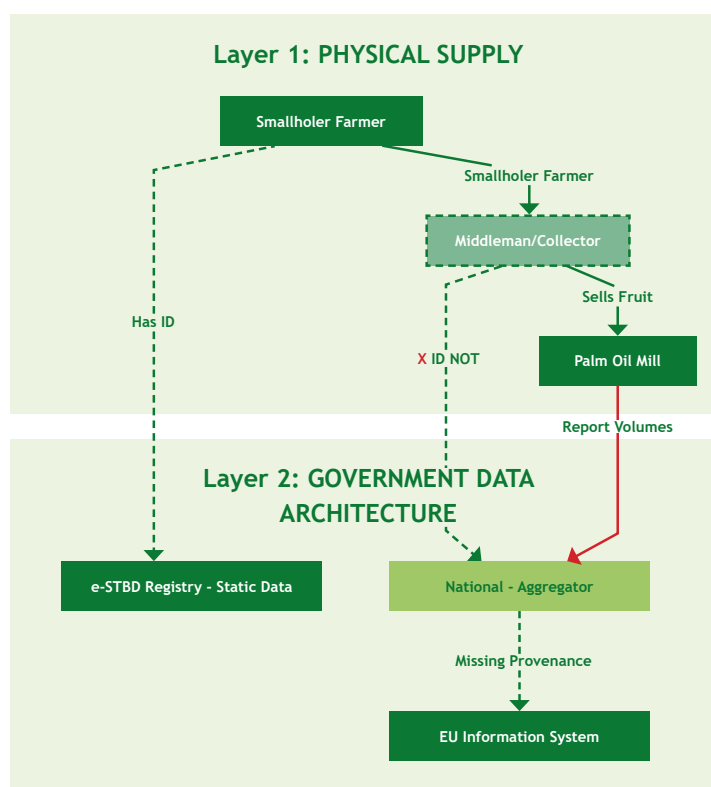
Transactions across dynamic supply chains pose a major data registration challenge in complying with future EUDR. A structured approach is essential to address the diversity of smallholders—from SMEs to large farmers (in some cases, e.g. in RSPO definition may be called as 'out growers')—and their unequal readiness levels. As explained before, these gaps are further widened by fragmented traceability systems, limited e-STDB registration, and digital infrastructure deficits. Consequently, generating Due Diligence Statements (DDS) for first-mile farmers remains highly complex for EU traders. The following matrix illustrates how these smallholder typologies align with EU operator mandates.

⁶ Coordinating Minister for Economic Affairs Decree Number 178 of 2024 (Keputusan Menteri Koordinator Bidang Perekonomian Nomor 178 Tahun 2024) concerning the Steering Committee for the Indonesia National Dashboard for Sustainable Commodity Data and Information

Table 2: Gap Analysis of EUDR requirement for operators/traders versus Indonesian system

EUDR Requirement	Indonesian System Component	Function	Status	Reason
1. Geolocation	e-STDB (Smallholders) SIPERIBUN (Corporates)	Repository of plot polygons and coordinates. Reason:	Fragmented	e-STDB holds the critical polygon data but acts as a static registry, not a live tracking tool.
2. Legality	SI-ISPO(Certification) e-STDB(Registration)	Verifies land tenure (SHM/SKT)* and operational licenses. Reason: *SHM- Land Ownership Certificate SKT _ Land ownership issued by the head of the village	Bottleneck	The administrative burden of land titling slows e-STDB registration, creating a barrier to entry.
3. Traceability (CoC)	<i>Private Sector Systems</i> (Currently) National Dashboard (Planned)	Tracking the flow of Fresh Fruit Bunches (FFB) from plot to mill.	Major Gap	The government systems (e-STDB) do not yet track volumes. Traceability currently relies on a private system which private companies establish by themselves to comply with market requirements, while to facilitate smallholder integration into the value chain systems, a government-led system was developed to support.

Source: Synthesized by the author (2024-2025). Technical data points adapted from EU Regulation 2023/1115, Kepmenko Perekonomian No. 178/2024, and Directorate General of Plantations data (2024)

Figure 2 - The Indonesia Gap (The middle section problem)

Source: Synthesized by the author (2026); based on technical gap analysis from Regional Technical Dialogues (RTD) #1-4 (2024-2025).

The Malaysian Model: Centralized State Architecture

In contrast to Indonesia, Malaysia's architecture is designed as a centralized national framework intended to facilitate closer coordination between public and private data. Under the proposed Sistem Kebolehejakan Nasional (SKN)—slated for phased implementation following discussions at PIPOC 2025—the registry, geospatial, and transaction layers are intended for integration under a unified approach. This creates a strategic digital backbone for compliance, moving beyond fragmented data silos toward a more cohesive national traceability framework as the system continues to mature.

Under the guidance of the Malaysian Palm Oil Board (MPOB), the sector operates on an alignment of three core pillars: the MSPO 2.0 regulatory standard (managed by the MPOCC), the SIMS transaction system, and the GeoSawit platform. While these platforms are managed by different agencies,

achieving interoperability is intended to support more consistent mass-balance checks, where reported trade volumes can be reconciled against the production capacity of verified legal boundaries. This mechanism is designed to enhance verification of both legality and deforestation-free status, with the objective of providing exporters a reliable foundation for meeting global compliance requirements.

Parallel to this, the Malaysian Rubber Board (LGM) enhances a similar centralized architecture to address the fragmentation inherent to rubber smallholders. The system is anchored by the PAT-G Card (Rubber Transaction Authority Permit), intended as a key ID mechanism for all rubber farmers. This ID feeds into RRIMniaga, a proprietary digital platform used by licensed dealers to record transactions at the first point of sale. When a smallholder sells cup lump or latex, the transaction is digitally recorded in real-time within RRIMniaga, linking the seller's ID directly to the traded volume. These transaction records are then cross-referenced against GeoRubber, the LGM's spatial inventory, to validate the origin.

⁷ Reference the PIPOC 2025 Keynote/Ministerial Address regarding the "Digital Backbone for EUDR Readiness.

⁸ Ministry of Plantation and Commodities (KPK). (2026). National Traceability System (SKN) Framework for EUDR Compliance: Integrating SIMS, GeoSawit, and e-MSPO. Kuala Lumpur: Government of Malaysia.

Within this proposed architecture, the state retains custody of the master data, while private-sector dealers and traders act as the operational interface. Currently, first-mile data is effectively locked within dealer networks, where transactional layers like SIMS and RRIMniaga capture daily trade. By interlocking these

private transactions with the state's geospatial layers (GeoSawit and RRIM GeoRubber), Malaysia establishes an unbroken chain of custody. This synergy allows EU operators to trace products back to state-verified origins, utilizing dealer-captured data to provide a robust defence against EUDR deforestation claims.

Table 2: Malaysia Palm Oil and Rubber Traceability Architecture

Layer	Palm Oil Architecture	Rubber Architecture
Regulator (The Hub)	MPOB (Malaysian Palm Oil Board)	LGM (Malaysian Rubber Board)
The User ID/License	MPOB License	PAT-G (Permit for Smallholders)
Transaction System	SIMS (Sawit Intelligent Management System)	RRIMniaga (Merchant/Dealer App)
Geospatial Database	GeoSawit	RRIM GeoRubber
National Standard	MSPO (MS 2530:2022)	MSNR (Malaysian Sustainable Natural Rubber)
Output	Traceable Export Certificate	Traceable Export Certificate

Table 3: Gap Analysis of EUDR requirement Indonesia and Malaysian System

EUDR Requirement	Malaysian System Component	Function & Status
1. Geolocation	GeoSawit (MPOB)	Function: A centralized GIS database containing polygons for all registered smallholders and estates. Status: <i>Scaling</i> . ⁹ While it serves as the national reference, efforts continue to cover gaps for independent smallholders (ISH).
2. Legality	MSPO (Mandatory Certification)	Function: Verifies legal compliance (land title, labour laws) via mandatory audits managed by MSPO. Status: <i>High Readiness</i> . ¹⁰ Legality is a prerequisite for the MSPO license to sell. However, gaps remain for smallholders currently in the certification transition phase
3. Traceability (CoC)	SIMS (Sawit Intelligent Management System); e-MSPO	Function: A digital ledger recording FFB transactions from farm to mill. Status: <i>Operational</i> . Links the volume sold to the specific GeoSawit ID, ensuring chain of custody.
4. Deforestation-Free	GeoSawit / MSPO Trace	Function: Integrated spatial verification between plot polygons and National Forest Inventory (NFI) baseline, cross-referenced with satellite imagery to verify the Dec 2020 cut-off date. Status: <i>High Readiness</i> . The system generates a “Green/Red” status for batches based on this overlay.

Source: Synthesized by the author (2026). Data grounded in Ministry of Plantation and Commodities (KPK) updates (Feb 2026) and the SKN Roadmap for EUDR enforcement.

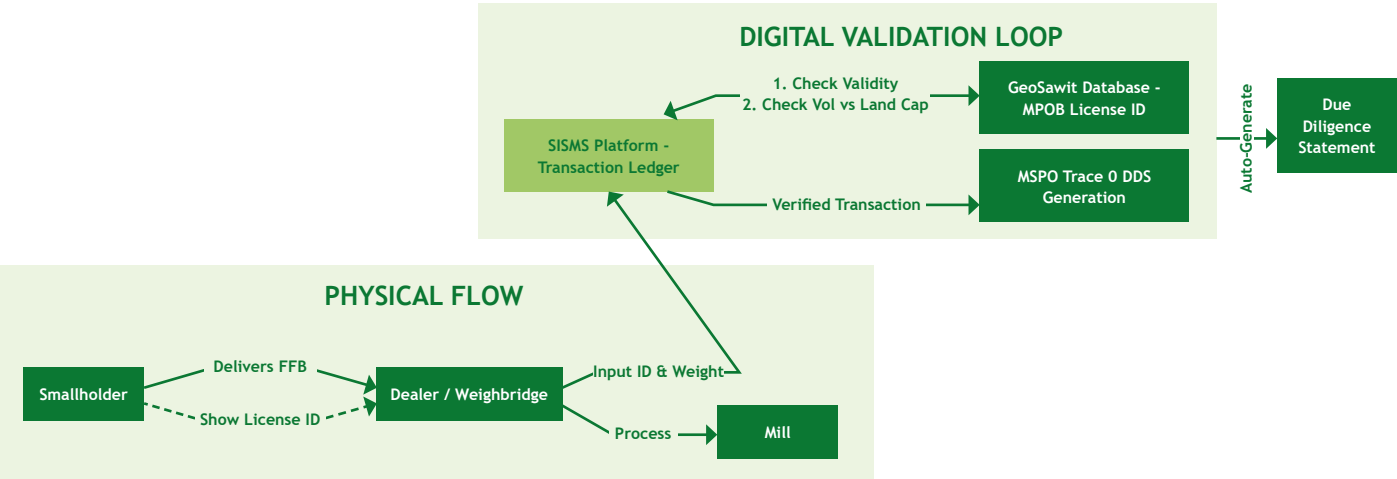
⁹ Sari, I. L., Weston, C. J., Newnham, G. J., & Volkova, L. (2022). Using Bayesian multitemporal classification to monitor tropical forest cover changes in Kalimantan, Indonesia. *International Journal of Digital Earth*, 15(1), 2061-2077. <https://doi.org/10.1080/17538947.2022.2146219>

¹⁰ Mohd Hanafiah, K., Abd Mutalib, A. H., Miard, P., Goh, C. S., Mohd Sah, S. A., & Ruppert, N. (2021). Impact of Malaysian palm oil on sustainable development goals: co-benefits and trade-offs across mitigation strategies. *Sustainability Science*, 17(4), 1639-1661. <https://doi.org/10.1007/s11625-021-01052-4>

The convenient interoperability across SIMS - Geosawit - e-MSPO - and DDS systems will enable traders to submit their Due Diligence Statement (DDS) and operators to check and verify. Unlike Indonesia, the Malaysian

system links the physical product to the digital ID at the point of transaction (the Mill Weighbridge) using the SIMS platform.

Figure 3 - The Integrated Malaysian Pipeline



Source: Synthesized and processed by the author (2026). Logic derived from the technical integration of MPOB’s SIMS (transactional), GeoSawit (geospatial), and the MSPO Trace (regulatory) frameworks as established under the National Traceability System (SKN) Roadmap (KPK, 2026)



Source: iStock

Table 4: Gap Analysis of EUDR requirement versus Malaysian system

EUDR Benchmark	ID Indonesian Palm Oil (e-STDB + SI ISPO)	MY Malaysian Palm Oil (e-MSPO, GeoSawit SIMS)	ID Cacao & Rubber	MY Cacao & Rubber (e.g., MSNR, MSNB Trace)
1. Geolocation Capability	LOW / FRAGMENTED eSTDB is designed for this (HIGH). But SI ISPO does not yet have capability, and the systems are not linked, making the data not yet usable at scale.	HIGH GeoSawit is a unified database designed to hold plot geolocation as a core component of certification.	EMERGING (multi crop e-STDB) No public system for this yet. The burden is 100% on private companies to build their own proprietary polygon databases.	EMERGING (HIGH) MSNR Trace provides end-to-end supply chain tracking for rubber, from plantation to final product. Meanwhile, cocoa is in transition via the National Cocoa Traceability System (NCTS); as of mid-2025, it covers 40% of domestic production and is fully aligned with the Malaysian Standard Premium Cocoa (MSPC) framework
2. Traceability (CoC)	NONE YET Neither e-STDB nor SI ISPO is a transactional system to allow traceability. They cannot yet segregate and link a CPO batch to a plot.	HIGH e-MSPO and SIMS is a national, digital traceability system built for exact purpose. MPOB licensing and database also allows traceability through each transaction.	EMERGING No public traceability system. All traceability is managed through private, proprietary systems (e.g., Olam's AtSource).	EMERGING MSNR Trace is designed to be the national traceability system for rubber, leveraging the PAT-G card for smallholder transactions.
3. Legality Verification	LOW / FRAGMENTED e-STDB's purpose is this, but there remained bottleneck (e.g., land title and land function prerequisites). ¹¹ SI ISPO verifies ISPO compliance.	HIGH Legality is a core principle of the MSPO standard, which is audited and managed within the e-MSPO system. It's all in one place. MPOB licensing also allows this.	NONE YET Legality is not yet managed at the national level. It's verified only by private certifiers (e.g., Rainforest Alliance) or by companies' own systems.	MEDIUM The Malaysian Rubber Board (LGM) license (PAT-G card) provides a strong baseline for smallholder legality and identification.
4. Small-holder Integration	LOW (DIFFICULT) The e-STDB legal-approach is a major barrier.	HIGH (EFFICIENT) The SPOC (Sustainable Palm Oil Cluster) model is far more efficient. It uses a group manager to onboard smallholders, solving the scalability problem.	VERY LOW Smallholders are not ready to be integrated. They are only integrated if a large buyer (e.g., Cargill) chooses to map them for their own private system.	HIGH Malaysia has a huge head start. The PAT-Gcard system for rubber smallholders already exists and has over 500,000 registered users.
5. Inter-operability & Data Output	VERY LOW Both are closed government portals and legality verified through maps, ownership and land-function.	MEDIUM It is a centralized, currently scaling toward coverage.	(NOT APPLICABLE) The private systems (e.g., MultiTrace) are interoperable under circumstances (payment) but they are not public systems.	MEDIUM It will follow the e-MSPO model: a centralized system where the government (LGM/MSPO) acts as the single data broker.
6. Deforestation-Free Monitoring	LOW The e-STDB plot data could be used for this, but as it's not linked to the supply chain, it's challenging to monitor products and only plots that can be monitored.	HIGH Compliance is anchored by GeoSawit, which maps estates and smallholder polygons. Because all certified plots are housed in the e-MSPO and GeoSawit national databases, it is possible to overlay satellite data to monitor the entire certified supply base.	NONE YET No public system. A company can only monitor the smallholders it has mapped in its private system. There is no sector-wide view.	EMERGING (HIGH) Once MSNR Trace is populated, it will have the same high monitoring.

Source: Synthesized and graded by the author (2026). Based on cross-sectoral analysis of MPOB, LGM, and MCB (Malaysia) data versus Kemenko Perekonomian and Ministry of Agriculture (Indonesia) technical roadmaps.

¹¹ There are different land ownership tiltes, SHM, HGU, HGB and different land functions, such as APL - other purposes areas, and so forth.

Source: iStock

Key Findings:

Barriers and Breakthroughs for Smallholder Inclusion

The transition toward a fully traceable supply chain is currently obstructed by a triad of deep-seated challenges: fragmented technical silos, a persistent digital divide, and the lack of viable financing models for low-margin commodities. These gaps manifest as disconnected, company-led data systems that create reporting fatigue for producers,¹² while complex intermediary chains often decouple the physical product from its geospatial origin.

While the palm oil sector benefits from established corporate infrastructure, the rubber and cacao sectors face a structural investment gap. To bridge this, the transition must shift from isolated pilots toward systemic ecosystems. This involves transforming traceability from a private administrative system into a public utility by leveraging Digital Public Goods (DPGs), aligning compliance costs with commodity-specific business models, and adopting landscape-level aggregation for scalability purpose.

The proposed transformation was synthesised to the context and methodology that was synthesized from the Regional Technical Dialogue (RTD #4)¹³ supplemented

by insights from prior sessions and current industry publications. This framework represents a proposed strategic model towards inclusive EUDR readiness and is not yet a fully implemented national system across all commodities.

Barrier and Breakthrough

Systemic Gaps in Cacao and Rubber and Breakthrough in Technology as a Public Good

The operational complexity of smallholder inclusion is best illustrated by the commodity-specific constraints shared by major industry players during the technical sessions of RTD #4 on traceability.

- Kirana Megatara (Rubber)¹⁴: As the largest crumb rubber producer in Indonesia, the company faces the logistical challenge of mapping hundreds of thousands of independent smallholders who contribute approximately 90% of its raw material. To achieve its target of 100% traceability by 2030, Kirana has deployed Rubber Notes¹⁵, a proprietary

¹² FGD Report

¹³ Synthesized from the Regional Technical Dialogue (RTD #4), “Strengthening Practical Traceability Solutions for Compliance and Market Competitiveness,” held on September 25, 2025. This dialogue convened multi-stakeholder experts from Indonesia and Malaysia, with technical insights provided by the European Commission Joint Research Centre (JRC) and the FAO to define operational breakthroughs for smallholder inclusion under the EUDR.

¹⁴ Synthesized from case studies presented during Regional Technical Dialogue #4, 2025

¹⁵ Sustainability Report 2023: Bridging the First Mile in Natural Rubber Traceability

application used by Sourcing Development Officers (SDOs) to track flow from farmgate to factory. In preparation for EUDR, the company has transitioned from sub-district tracking to site-specific polygon mapping via the RubberWay risk-mapping tool. Despite these tools, the first mile remains structurally complex; because rubber is traded through multi-tiered intermediary chains, data continuity often stalls at the dealer level. To bridge this, Kirana Megatara collaborates on the CASCADE¹⁶ project (alongside Michelin and Porsche), utilizing the SUTTI and RubberWay apps to provide agronomy training and financial literacy. These initiatives aim to trade knowledge for data, yet scaling such high-touch models remains a significant administrative and financial hurdle.

- Olam Food Ingredients / OFI (Cocoa): Mapping the Indonesian cocoa supply chain is a massive undertaking due to fragmented plots, informal land tenure, and the dominance of independent smallholders. As a pioneer in digitalization since 2017, OFI has implemented the OFI Direct system¹⁷ to record farmer identities, village locations, tree counts, and production estimates. Moving beyond single GPS points, OFI has transitioned to 100% polygon mapping¹⁸ to meet global buyer demands and prepare for EUDR compliance across both certified and conventional products. However, field effectiveness relies heavily on data synchronization from partner traders. Systemic challenges arise from digital infrastructure gaps, where limited internet connectivity in remote areas of Sumatra and Sulawesi often obstructs real-time syncing. These challenges are further aggravated by the region's archipelagic geography, where inconsistent cellular coverage often renders real-time data tools ineffective.

Furthermore, farmer economic behaviour remains a hurdle; many prefer selling to mobile traders for immediate cash without the requirements of quality standards or registration. Without a push factor such as clear regulatory mandates or direct price incentives, the urgency for farmers to participate remains low¹⁹.

To overcome connectivity barriers, Regional Technical Dialogue (RTD) #4 highlighted Digital Public Goods (DPGs) as offline solutions. A key operational example is Open Foris Ground²⁰, which empowers smallholders to map farm boundaries in remote areas without a cellular signal, synchronizing polygons once connectivity is restored to ensure a seamless digital chain of custody.

The data is then processed by the FAO's WHISP (What Is in that Plot)²⁰ engine, which functions as an automated validation layer by combining global, regional, and national datasets to assess deforestation risks. To ensure high-fidelity monitoring, the system integrates the IMPACT Toolbox for intersecting geolocation data with forest maps and utilizes the JRC Global Forest Cover Map 2020²² as a consistent global reference layer for risk assessment. By layering these datasets over existing state-owned assets—such as Indonesia's SIMONTANA and the One Map Policy—the system creates a National Validation Gate. This architecture is designed to expedite the issuance of e-STDB registrations at scale by flagging regional risks across millions of plots simultaneously through a high-resolution forest baseline.²³

In Malaysia, the strategy focuses on operationalizing the newly integrated National Traceability System (SKN). This platform—designed to bridge SIMS, GeoSawit, and e-MSPO—is currently being implemented to provide exporters with a consolidated path to verified compliance. For the rubber sector, the linkage between the PAT-G card and the RRMniaga 2.0 app facilitates a process where trade volumes are recorded against verified polygons at the point of primary sale. This initiative serves to digitize smallholder transactions²⁴ within the national supply chain framework as the system scales toward full interoperability.

This architecture resolves data privacy friction by utilizing the national validation system to convert sensitive GIS data into a binary compliance token via API. This provides verifiable proof for EUDR Due Diligence (DDS) without violating the data provisions of Law 39/2014, ensuring data is used by competent authorities for compliance checks without being made public.

¹⁶ Michelin, Accessed February 10, 2026, <https://natural-rubber.michelin.com/projet-cascade-caoutchouc-naturel-durable-michelin>

¹⁷ Ofi (olam food ingredients). (2024). Cocoa Compass: Performance Report 2023/24 - Navigating the Future of Sustainable Cocoa. London: ofi

¹⁸ Cocoa Compass: Impact Report 2023/24 - Navigating the Future of Sustainable Cocoa. Available at: ofi.com/sustainability. Note: ofi achieved 100% traceability to farm/community in 2020; transition to full polygon mapping reached 79% by mid-2024.

¹⁹ Synthesized from interview and case studies presented during Regional Technical Dialogue #4, 2025

²⁰ Open Foris Ground: Food and Agriculture Organization of the United Nations (FAO). (2024). *Open Foris: Innovative tools for environmental monitoring*

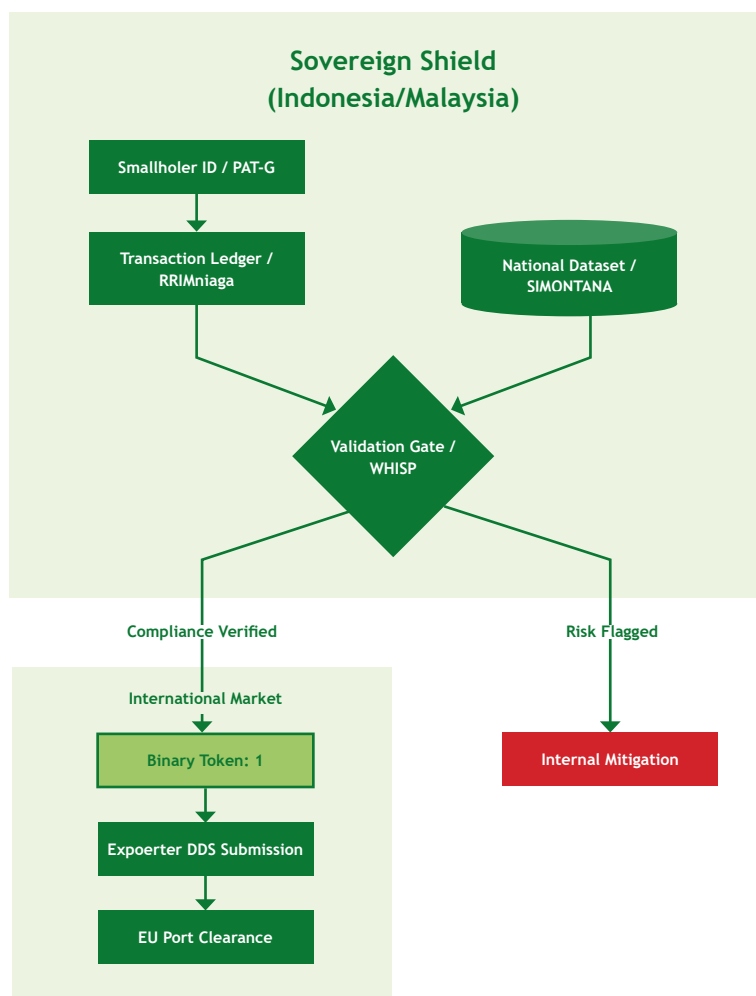
²¹ WHISP (What Is in that Plot): World Resources Institute (WRI) & FAO. (2025). *WHISP: An automated validation engine for EUDR geospatial compliance*.

²² JRC Global Forest Cover Map & IMPACT Toolbox: European Commission Joint Research Centre (JRC). (2025). *Geospatial tools for EUDR Risk Assessment*.

²³ Sovereign Traceability: Tropical Forest Alliance (TFA). (2026). *Regional Technical Dialogue #4 Summary Report: Inclusive Traceability for Independent Smallholders*.

²⁴ Shamsudin, M., & Rahman, A. (2025). "Digitalizing the Smallholder: Integrating Mobile Applications with National Licensing Systems in the Malaysian Rubber Industry." *Journal of Tropical Agriculture and Sustainability*, 12(1), 45-58.

Figure 2 - The National Validation Gate - Automating EUDR compliance through sovereign integration and binary tokenization.



Source: Synthesised by the author based on the RTD #4 discussion

Financial Viability Gap & Breakthrough: Aligning Business Models and Margins

Even when technology functions perfectly, traceability incurs costs—typically estimated at \$15-\$20²⁵ per smallholder for initial mapping and verification according to discussions held during FGD. If the cost of data collection outweighs the crop's profit margin, the system becomes unsustainable; it risks collapsing the moment project-based initiatives conclude. In the absence of off-takers willing to front-load compliance costs, the market naturally pivots toward larger estates that can absorb these overheads that would be challenges for independent producers for market inclusion.

The breakthrough lies in aligning business models with the economic realities of specific commodities:

- **Premium Model (Cocoa):** In the cocoa sector, such as the partnership between the Kerta Samaya Samaya (KSS) Cooperative and Valrhona in Bali²⁶, compliance is subsidized by quality. The premium price for high-quality fermented beans funds the Internal Control System (ICS), effectively embedding compliance costs into the final product value.
- **Knowledge-for-Data Model (Rubber)**²⁷: In the rubber sector, companies like Kirana Megatara trade knowledge for data. By leveraging agronomy teams to map farm boundaries while providing technical training on Good Agricultural Practices

²⁵ Taken from the FGD discussion of smallholders and companies during the inception phase

²⁶ Sector-Specific Examples: Pages 14 and 15 highlight the KSS Bali cooperative and the mapping struggles in Musi Banyuasin.

²⁷ Regional Technical Dialogue #4 Page 11-12: Realities of company-driven mapping in rubber supply chains.

(GAP), farmers exchange geolocation data for yield-boosting support. This allows the company to manage complex deforestation analysis centrally while addressing resource-intensive mapping challenges.

- **Supply Shed Model (Palm Oil):**²⁸ For the palm oil sector, the Musim Mas approach treats traceability as a necessary procurement investment to secure a steady flow of compliant Fresh Fruit Bunches (FFB). By absorbing mapping and STDB (Surat Tanda Daftar Budidaya) facilitation costs for smallholders within a specific mill radius, the company de-risks its entire supply base. This shifts the incentive from a simple price premium to long-term market security, ensuring the supply remains “market-ready” for the EU.

Scalability and Aggregation Gap & Breakthrough: Systemic Solutions/Innovations at Scale for Independent Smallholders

To move beyond isolated pilots, the commodity sector requires structural innovations that lower the per-farm mapping costs and shift the companies’ verification process for farmer’s data collection toward aggregated entities, i.e. cooperatives or intermediaries, in both Indonesia and Malaysia. This scalability gap is further widened by land tenurial complexities, which remain the most critical barriers to progress. Without accelerated government action on land titling and the establishment of clear legal pathways, evolving regulatory frameworks risk widening the integration gap for smallholders, rather than fostering participation.

A. Verified Sourcing Areas & Jurisdictional-Led Model

The breakthrough for smallholder inclusion involves a strategic shift toward regional aggregation via the Verified Sourcing Areas (VSA) model.²⁹ Rather

than a singular, static system, this represents a series of targeted interoperability layers currently being operationalized to bridge the gap between landscape-level monitoring and individual plot requirements.

Under this framework, District Governments (Dinas Perkebunan) lead the mapping and verification of smallholders within their administrative boundaries—supported by national budgets such as BDPKPS—to fulfill mandatory e-STDB registration. While the One Map Policy serves as the Government’s Master Spatial Database for resolving land-tenurial legality and forest boundaries, the proposed National Dashboard acts as the interoperability gate for external markets.

However, the use of jurisdictional evidence does not absolve private actors of their legal mandates. Operators placing products on the EU market remain ultimately responsible for assessing and mitigating risks under the EUDR. They must verify the jurisdiction’s mapping methodology and ensure that the forest definitions used (e.g., via SIMONTANA) align strictly with EUDR standards.

This middle-ground architecture offers a potential pathway for private traceability platforms to cross-reference coordinates against District-verified polygons via API, upholding national data sovereignty by issuing binary compliance tokens (Green/Red) instead of raw sensitive data. This approach—currently being piloted and tested within the PPI Compact in Aceh Tamiang—seeks to de-risk the ‘first mile’ by aggregating smallholders into digitally tracked, investable clusters. By providing a framework for validated evidence, such a system could support an operator’s obligation to fulfill the requirements of a Due Diligence Statement (DDS), offering a scalable model for high-fidelity compliance data.

²⁸ Regional Technical Dialogue #4: Private Sector “Supply Shed” Realities: Pages 11-13 outline how company-led mapping secures market access and integrates with national systems.

²⁹ <https://sourceup.org>

B. SPOC Model

The organizational aggregation through the SPOC Model in Malaysia provides a foundation for compliance, though it must be framed correctly within the EUDR context³⁰. While the EU does not view state or voluntary certifications as a green lane for compliance, these systems serve as a national data backbone.

The role of the MPOB and LGM is not to replace the operator's due diligence, but to act as a verified data provider. For palm oil, the MPOB manages data collection at scale through 162 SPOCs to centralize the mapping and verification of 260,000 independent smallholders. At the point of sale, the e-MSPO, SIMS, and RRIMniaga systems provide the geospatial evidence and legal documentation for a Due Diligence Statement (DDS). This interoperable infrastructure ensures individual plot data is accessible and verifiable, streamlining global market compliance.

In Indonesia, this architecture utilizes a validation bridge to solve transparency and data privacy. Instead of mandating the disclosure of raw, sensitive ownership data protected by Plantation Law 39/2014, the national system issues a Binary Compliance Token via API. This token confirms that a specific batch of commodity is tethered to a verified, deforestation-free polygon within the state's master database.

By integrating tools like WHISP, operators can independently verify these national tokens against global forest layers³¹. This hybrid approach respects national sovereignty while providing the high-granularity, verifiable evidence that the EU market demands. Ultimately, these national clusters and registries are the only scalable way to ensure that the of smallholders is not de-risked out of the global supply chain, but rather integrated through a transparent, digitally-verified architecture.

Operationalizing the Breakthroughs and their integration into the national traceability infrastructures and pipelines

The private sector and various initiatives identified above, have begun to play a critical role in filling operational and data gaps required by the EU Deforestation Regulation (EUDR). Large buyers, refiners, traders and consumer-goods companies - either themselves or via multi-stakeholder initiatives - are now deploying their own traceability platforms, satellite monitoring solutions, independent verification models and supplier-engagement programmes. Operationalising these private systems in alignment with Indonesian and Malaysian regulatory frameworks can accelerate national readiness, specifically to facilitate independent smallholder integration, and reduce the burden on government institutions.

While previous sections highlighted necessary economic incentives, this analysis concludes that their efficacy relies on successful integration into technical pipelines. Synthesizing the Roundtable Discussions (RTD), the following strategic functional design is proposed to resolve architectural infrastructures within the Indonesian and Malaysian systems, specifically addressing traceability context.

Indonesia Perspective: Closing the Connectivity

In the Indonesian context, the critical challenge in the supply chain is the disconnection between Layer 1 (The Middleman/Aggregator) and Layer 2 (The National Infrastructure). Currently, while the physical product moves seamlessly from farm to mill, the digital identity—such as the e-STDB ID—is frequently lost at the first point of aggregation. By utilizing interoperable, open-source standards, this approach aims to ensure that traceability data remains tethered to the physical product as it moves through various

³⁰ Operational breakthrough in jurisdictional-scale verification; demonstrated via technical slides during Regional Technical Dialogue #4 (2025)

³¹ EUDR Geospatial Baseline: Bourgoïn, C., et al. (2025). Global forest maps for the year 2020 to support the EU regulation on deforestation-free supply chains. Publications Office of the European Union.

levels of aggregation, regardless of the ultimate architecture of the national data system. According to the sessions facilitated in the RTD #4, some models had been piloted and operationalised and explored its potentiality further.

A. Shifting from Anonymous Collection to Verified Aggregation

The breakthrough in the first mile involves transitioning from informal, anonymous collection toward a model of Verified Aggregation. Rather than allowing middlemen to strip data from the supply chain, this model operationalizes the collector as a Data Conduit.

In functional pilot areas like Aceh Tamiang, this transition is managed through a Segregated Supply Chain involving cooperatives and certified mills, such as PT Mora Niaga Jaya. To ensure a concrete link between the product and the plot—addressing the EUDR’s core requirement—the following operational layers are used:

Point-of-Sale Data Capture: Instead of anonymous cash transactions, the Verified Aggregator uses the Jejak Sawit app to record the Farmer’s Name, NIK (National ID), Polygon coordinates, and e-STDB registration number at the moment of collection.

Physical Safeguards: To prevent leakage or the laundering of unverified fruit, FFB is weighed at specific pick-up points (RAMPS) and cross-checked again at the mill gate to ensure the physical volume matches the registered production quota of the verified polygons.

Digital Integrity: The system utilizes Live Truck Tracking to monitor the journey from the farm to the mill, providing a secure Digital ID for the batch.

B. Scaling via the Jurisdictional Umbrella and DPGs

Complementing the aggregator approach is the Verified Sourcing Area (VSA) model, which functions as a collective risk mitigation approach designed to address impacts of commodity production that are beyond the full control of individual companies. In the context of the EUDR, this Jurisdictional Approach (JA) is defined by political-administrative boundaries, such as districts, and is used to coordinate policies, enforcement, and investments at scale. Rather than verifying every

transaction in isolation, this model convenes stakeholders—including governments, companies, and civil society—to collaboratively pursue shared goals through a joint roadmap to minimize social and environmental conflicts.

Operational evidence of this model is seen in the PPI Compact in Aceh Tamiang, a multi-stakeholder partnership (MSP) that has already accelerated the certification of 2,200 smallholders. Within this framework, the system leverages the Jejak Sawit app to bridge the middleware gap. The app captures live data—including Farmer Name, NIK, Polygon coordinates, and e-STDB numbers—to link the physical flow of goods to verified plots.

While the JA serves as risk mitigation tool (Article 11), it does not absolve private actors of their legal mandates. Operators placing products on the EU market remain ultimately responsible for the entire three-step due diligence process, including the requirement for precise plot-level geolocation for every batch.

Malaysia Perspective: Optimizing the Dealer Node

In the Malaysian context (Figure 3), the pipeline is already technically connected via the MPOB License ID. The operational challenge is not a broken link, but rather a bottleneck at the Dealer Node, where thousands of smallholders converge to sell their harvest.

The SPOC model requires every independent smallholder to have its own GeoSawit data (managed by MPOB) alongside their MPOB License compliance may create significant friction at the dealer level. While the Sustainable Palm Oil Cluster (SPOC) is not designed as a dealer-level or digital system, its group certification structure can reduce operational complexity for downstream actors by centralising smallholder administration and compliance coordination through a SPOC Group Manager. The Group Manager oversees the Internal Control System (ICS) and may utilise digital tools to facilitate data management and reporting, without altering the fundamentally physical and institutional nature of the SPOC.

Where digital reporting systems such as SIMS are used, the SPOC Group Manager may assist smallholders by

coordinating information required for submission and ensuring records are complete and consistent at the group level. This administrative support helps reduce the likelihood of delivery disruptions at the mill due to documentation issues. Responsibility for system access, licence status, and final acceptance remains with the relevant MPOB and mill processes, while the SPOC's role is limited to organisational coordination under the Internal Control System (ICS). In this case, SIMS entry is managed by dealer/collection centre/ramps (for smallholder data), and can be coordinated with the SPOC group manager.

Malaysian rubber has their own system - MSNR Trace³²- where for the Malaysian rubber sector, Malaysia

is integrating into the government architecture. This integration leverages the existing PAT-G transaction card—already held by over 500,000 rubber smallholders—as the primary Digital ID within Figure 2. By upgrading card readers at the dealer level to connect with the MSNR Trace system, the operational flow becomes automated: rubber cannot be sold to a factory unless the PAT-G card is swiped. This action automatically links the trade volume to the farmer's polygon in the national database, creating a seamless digital audit trail parallel to the physical goods.

Table 5: Operationalizing Traceability: From bottlenecks to breakthrough

Market	Bottleneck	Operational Solution	Technical Enabler (tool)
Indonesia (Palm & Cocoa)	The middleware gap Digital IDs are lost at the first point of sale due to anonymous middlemen.	<u>The verified aggregator</u> Shift procurement to formal entities who mandate ID capture at the transaction point.	<u>Integrated Traceability</u> ODK/Corporate Apps linked to e-STDB & AI-enhanced satellite filtering for shade-grown plots.
Indonesia (Systemic)	<u>The Verification Ceiling</u> Verifying every single plot in real-time creates data congestion and trade delays.	<u>The VSA Model</u> Adopt a “Jurisdictional Umbrella” as a collective Risk Mitigation Measure (Art. 11) to validate sourcing areas	<u>Origin Proxy Validation</u> (One Map Policy / Dashboard Layer 2)
Malaysia (Palm Oil)	<u>The Dealer Node Friction</u> Independent smallholders lack the technical capacity for data entry, slowing trade.	<u>Administrative Proxy</u> SPOC Managers assist smallholders with digital tools to facilitate data sharing with aggregators like SIMS	<u>Proxy Data Entry</u> SIMS Access for Dealers & SPOC-led data uploads.
Malaysia (Rubber)	<u>The Data Vacuum</u> Rubber supply chains lack the mature digital pipelines of the palm oil sector.	<u>The Resilience Model</u> Utilize existing transaction cards to force-link volume to polygons at the weighbridge.	<u>Synchronized Linkage</u> (PAT-G Card + MSNR Trace Readers)

Source: Synthesized by the author (2026). Technical solutions adapted from presentation slides and case studies (Kirana Megatara, MPOB, IDH) shared during Regional Technical Dialogue #4 (2025)

³² Malaysian Rubber Board (LGM). (2025, June 24). Launch of MSNR Trace: Digital Platform for Sustainable Natural Rubber Supply Chain Traceability. Kuala Lumpur: LGM.n.

Source: Shutterstock

The Strategic Recommendation

This chapter provides a synthesis of the Roundtable Discussion (RTD) and an analytical framework for operationalizing compliance through existing national infrastructure. By bridging technical pipelines with strategic policy, the following recommendations for government, private sector actors, and smallholders/cooperatives are designed to accelerate EUDR readiness. The objective is to strengthen smallholder inclusion and eliminate system redundancies, transforming traceability from a resource-intensive requirement into a streamlined, high-integrity digital asset.

Government (Indonesia and Malaysia): Enabling an Interoperable Ecosystem

The priority is to transition from fragmented projects to interoperable national data architectures. State institutions—such as the Coordinating Ministry for Economic Affairs and the Ministry of Agriculture—are responsible for maintaining the standardized reference datasets and protocols that allow private operators to fulfill their individual Due Diligence obligations.

Operationalizing EUDR-Aligned Risk Mitigation

Governments should formalize a national risk protocol based on deforestation risk (post-2020), legality, and

monitoring capacity. While plot-level geolocation remains mandatory for every batch under EUDR, this jurisdictional classification functions as a robust risk-mitigation tool under Article 11. It provides the standardized and independently auditable framework needed to justify negligible risk findings during the due diligence process, ensuring that regional aggregation supports—rather than replaces—the individual operator's declaration.

Strengthening Data Interoperability & Access

Rather than a centralized reporting interface, governments should focus on implementing API-based environments to facilitate data verification. This system bridges the gap between private sector due diligence and national sovereignty by allowing Operators to verify supply chain data against government-verified legality and forest cover records. This functions as a critical risk-mitigation layer, providing the necessary evidence for negligible risk findings while ensuring that the final due diligence responsibility remains with the private operator.

Dynamic Farmer Registries & Satellite Monitoring

The shift from static snapshots to living registries (e.g., e-STDB) must be integrated with real-time satellite detection. This ensures that the “spatial truth” used for verification is updated periodically,

reducing the re-registration burden on smallholders while strengthening the integrity of the audit trail over time.

Formalize Verified Aggregation

Governments should recognize and operationalize first-mile actors—collectors, dealers, and cooperatives—as formal verification points. By institutionalizing minimum data capture rules at the first point of sale (e.g., using functional tools like the Jejak Sawit app), governments can prevent identity loss during aggregation. This ensures smallholder volumes remain digitally linked to their pre-verified polygons, providing the machine-readable evidence (NIK, STDB, and coordinates) required for a valid Due Diligence Statement (DDS).

Private Sector: Moving from Documentation to Operations

Private sector actors carry the direct legal burden for compliance. Due diligence must be treated as a live operational system that ensures every batch is linked to a verified origin.

The Three-Layer Compliance Pipeline

Companies should structure their EUDR readiness across three integrated layers:

- **Identity & Origin:** Linking specific farm-level geolocations to physical volumes.
- **Integrity & Risk:** Implementing automated screening and mitigation for flagged plots.
- **Auditability:** Maintaining a “paperless” audit trail of traceable decisions and timestamped logs.

Securing the *First Mile*

Mapping efforts must prioritize the first-mile aggregation stage (collectors, dealers, and cooperatives), where data fragmentation is highest. Minimum records must be standardized at the point of first sale to ensure the farmer’s identity is preserved as goods move toward the mill.

Interoperability Over Proprietary Silos

To reduce reporting fatigue and costs, companies should adopt shared data standards. Where credible government-validated outputs or jurisdictional risk assessments are available, private systems should utilize them as a verification layer to reduce duplicative mapping and screening.

Incentives and Cooperative Partnerships

Compliance is only sustainable if it is financially viable for smallholders.

- **Incentive Structures:** Companies should bundle compliance with productivity packages, stable off-take agreements, or targeted premiums.
- **Structured Onboarding:** Engage cooperatives through Internal Control Systems (ICS) and remediation support to ensure smallholders are not excluded due to administrative gaps.

Smallholders & Cooperatives: Facilitated Inclusion

Smallholders are essential to inclusive compliance, but they require systematic education and technical assistance. Support agencies and partnering institutions must lead the transition to ensure compliance does not become a disproportionate cost burden for the “first mile.”

Accelerated Registration (e-STDB)

Enrolment in e-STDB is the baseline for identity and legality. Cooperatives and support agencies should coordinate mass registration drives to formalize smallholder recognition and unlock eligibility for wider compliance programs.

Facilitated Group Mapping & Polygon Capture

To achieve scale and reduce costs, mapping should be organized via cooperative or group-based models. Rather than mapping independently, smallholders should be supported by technical assistance units

(cooperatives, NGOs, or consultants) who provide mapping as a service to ensure data technicality meets EUDR standards while allowing farmers to maintain data ownership.

Integrity in Transaction Documentation

Cooperatives must implement standardized logs (plot reference, volume, date, buyer ID). This preserves the link between the origin and the physical volume during aggregation, preventing identity loss before the goods reach the mill.

Internal Control Systems (ICS) & Governance

Cooperatives should transition into compliance hubs by strengthening their ICS. This includes managing member registries, internal monitoring, and grievance handling—functions that build the “audit-readiness” required by downstream buyers.

Linking Compliance to Tangible Benefits

Compliance adoption is driven by survival and growth. Support agencies must link progress to:

- Secured Off-take: Guaranteed market access for compliant volumes.
- Service Bundling: Access to technical inputs, productivity training, and innovative financing.

To scale compliance, stakeholders must align on three enabling conditions that bridge the gap between smallholder reality and international requirements.

Cross Stakeholder Priority of Actions

Interoperability via State Leadership

To eliminate data silos, national and private systems must communicate through shared standards.

- Responsibility: State institutions are the primary leads. They must define the API protocols and technical language that allow private tools to verify data against national registries.
- Goal: Use government-validated outputs to support private Due Diligence, reducing duplicative mapping and costs.

Verified Aggregation as the Scalable Pathway

Formalizing collectors and cooperatives as “verification points” is the only way to prevent data loss in Indonesia’s fragmented supply chain.

- The Mechanism: This ensures farmer identity and geolocation remain digitally tethered to physical volumes during the aggregation process, satisfying the EUDR’s 1-to-1 traceability requirement without halting trade.

Integrated Finance & Capacity Building

Compliance must be facilitated by support agencies and funded through shared-responsibility models.

- Facilitated Mapping: Technical partners (NGOs, consultants) should manage geodata on behalf of farmers to ensure accuracy and scale.
- Risk Mitigation Tool: These collective efforts are presented not as a declaration in excess, but as a risk-mitigation tool. While operators remain legally responsible for their imports, sourcing from Verified Aggregation zones provides the evidence needed to present negligible risk.



Source: iStock